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Section: BSIT – 3209
Subject: BPM
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1. How do Data Analytics and BI Tools specifically assist an organization in identifying “opportunities for improvement” within their existing business processes?
-Data Analytics and BI tools help an organization find opportunities for improvement by looking at all the information from different parts of the business. For example, they can show exactly where a process is slow or where we are spending too much money. Instead of guessing, we can see the problem clearly in a chart or report. This helps us know exactly which step to fix to make the whole process better and faster

2. Infrastructure Evolution: Compare the traditional model of maintaining physical IT infrastructure with the benefits of Cloud Computing and Saas. Why is the latter considered more "scalable" for growing organizations?
-In the traditional model, a company had to buy its own servers and software, which took a lot of time and money to set up and maintain. With Cloud Computing and SaaS, you just use the software over the internet and pay for what you use. This is more scalable for a growing organization because you can easily get more storage or users immediately without waiting to buy and install new physical machines. If you shrink, you can also reduce what you pay for.

3. The text mentions that RPA “frees up human resources.” Explain the relationship between automating “rule-based tasks” and the ability of a workforce to focus on “higher-value activities.”
-The relationship is simple: when we use RPA to do boring, rule-based tasks like entering data from one form into a computer, it gives people back their time. Instead of spending hours on repetitive work, the workforce can now use their brains for higher-value activities. This means they can focus on things like solving customer problems, coming up with new ideas, or improving the business, which computers cannot do on their own.

4. Describe how IoT technology transforms physical sensors into actionable business intelligence. Provide one example of how this applies to “predictive maintenance.”
-IoT technology puts sensors on physical things like machines. These sensors collect real-time data, like temperature or vibration, and send it to a computer system. This raw data is then turned into business intelligence, which is useful information. For example, in predictive maintenance, an IoT sensor on a motor can detect that it is getting too hot. This information triggers an alert so we can fix it before it breaks down completely, saving time and money.

5. In the context of Blockchain, explain how a “decentralized and secure ledger” changes the way multiple organizations might share data compared to a traditional centralized database.
-With a traditional centralized database, one organization controls all the data. If other organizations want to share it, they have to trust that one source and worry about errors. Blockchain changes this because it is a decentralized and secure ledger. This means every organization in the network has their own copy of the exact same data. Any change must be agreed upon by everyone. This creates trust and transparency because no single person can change the records secretly.

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6. Artificial Intelligence is described as a tool for "decision-making." Explain how AI differs from standard software programming when it comes to "detecting patterns" in large datasets.

-Standard software programming follows a strict list of instructions that a programmer writes. It does exactly what it is told. AI is different for detecting patterns because it can learn from the data itself. Instead of being told "look for this specific thing," the AI can analyze a huge dataset and find patterns or trends that a human programmer might not even know to look for. It makes its own rules based on what it finds in the information.

7. Based on the Cybersecurity section, explain how tools like encryption and multi-factor authentication specifically support the "confidentiality" and "integrity" of organizational data.

-Encryption supports confidentiality by scrambling the data. If someone steals it, they cannot read it because it looks like nonsense. Multi-factor authentication also supports confidentiality by making sure the person trying to access the data is really who they say they are, like using a password and a code from their phone. For integrity, these tools help make sure the data is not changed by the wrong people. By controlling access, we can keep the data accurate and trustworthy.

8. Choose two technologies from the list (e.g., IoT and AI) and explain how they might work together to "optimize resource utilization" in a smart warehouse setting.

-In a smart warehouse, IoT and AI can work together to optimize resource use. IoT sensors on forklifts can track their location and if they are being used. This location data is sent to an AI system. The AI analyzes this information to see which forklifts are idle and which ones are in high demand. It can then automatically route tasks to the nearest free forklift, saving fuel and time and making sure all equipment is used efficiently.

9. How does Blockchain enhance "digital identity verification," and why is this critical for modern information systems?

-Blockchain enhances digital identity verification by creating a secure and unchangeable record of a person's identity. Instead of showing all your private information every time, you can use the blockchain to prove you are who you say you are without revealing everything. This is critical for modern information systems because identity theft and fraud are big problems. Blockchain makes it much harder for someone to fake an identity or steal one.

10. Using the description for AI and Machine Learning, explain the process of how "deriving insights" from customer data leads to a "personalized customer experience."

-AI and Machine Learning can process huge amounts of customer data, like what they buy or what websites they visit. By doing this, they derive insights, which means they find patterns, such as "customers who buy a tent also often buy a sleeping bag." The business then uses this insight to create a personalized customer experience. For example, when that customer visits the website next time, it might automatically show them sleeping bags or special deals on camping gear.